BTS-I(R/S)-11-23-2197 Reg. No. 20623042 **B**

B.Tech. Degree I Semester Regular Examination November 2023

CE/ME/SE 23-200-0106 A ENVIRONMENTAL AND LIFE SCIENCES

(2023 Scheme)

Time: 3 Hours Maximum Marks: 50

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Identify the global environmental issues.
CO2: Examine the types of pollution in society along with their sources.
CO3: Elucidate the basic biological concept via relevant industrial applications and case studies.
CO4: Evaluate the principles of design and development, for exploring novel bioengineering projects.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A
(Answer *ALL* questions)

		$(5 \times 2 = 10)$	Marks	BL	CO	PO
I.	(a)	Mention some consumptive uses of biodiversity.	2	L1	1	2
	(b)	What role does biodiversity play in the economic development of mankind?	2	L2	3	2
	(c)	Write a short note on the methods of land reclamation.	2	L2	2	2
	(d)	What is the role of pacemaker and defibrillator?	2	L4	4	2
	(e)	Explain in short, the functioning of lungs.	2	L3	4	2

PART B

		$(4 \times 10 = 40)$	Marks	BL	CO	PO
II.		Discuss the structure and function of an ecosystem.	10	L1	1	2
		OR				
III.		Explain the structure of a grassland ecosystem.	10	L1	1	2
IV.		Explain the various effluent treatment processes.	10	L2	2	2
		OR				
V.		Write a short note on the management of:				
	(i)	E-waste	5	L2	2	2
	(ii)	Drought.	5	L2	2	2
VI.		Mention the application of:				
	(i)	*Glucose oxidase in biosensor*	5	L3	3	2
	(ii)	*Lipids as cleansers.*	5	L3	3	2
		OR				
VII.		Write short notes on:				
	(i)	*Friction reducing shark skin*	5	L3	3	2
	(ii)	*Aerodynamic design of birds.*	5	L3	3	2
VIII.		Explain the working of				
	(i)	EEG	5	L4	4	2
	(ii)	ECG.	5	L4	4	2
		OR				
IX.		Discuss the application of various bio printing techniques for 3D printing of ears and food.	10	L4	4	2

Bloom's Taxonomy Levels

L1 - 24%, L2 - 26%, L3 - 25%, L4 - 25%.

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